

Finishing Pittsburgh's Sidewalk Accessibility Audit

Project Sidewalk's volunteers have hand-audited **334 of Pittsburgh's 828 walkable km** in six years. The city is writing **its first ADA Transition Plan in 30+ years**, which requires a barrier inventory. We extend the published curb-ramp detector to all five barrier types, run it over the un-audited 494 km, and hand the city a repair list ranked by pedestrian demand.

334 / 828 KM AUDITED · PITTSBURGH	25,690 HUMAN LABELS W/ SEVERITY + PEER VOTES	214,376 PRE-LABELLED PANORAMAS, IMAGES INCL.	~25k LABELLED CROPS ACROSS 5 BARRIER CLASSES	0.95 / 0.87 PUBLISHED P/R – CURB RAMPS ONLY
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THE GAP WE'RE FILLING

RampNet (Project Sidewalk, Aug 2025) detects curb ramps at 0.949 P / 0.873 R — but **only curb ramps**, trained only on NYC, Portland OR and Bend OR, the three cities with government curb-ramp inventories. Nobody has a model for the other four types, and nobody has run any of it on Pittsburgh. The paper's limitations name both, and explicitly invite work on “missing curb ramps, path obstacles, surface problems”.

WHAT WE BUILD

- Five-class detector.** Extend RampNet's heatmap-regression head from curb ramp to *missing* curb ramp, obstacle, surface problem and missing sidewalk, using the ~5,100 published crops per class.
- Severity head.** 20,453 Pittsburgh labels carry a 1–5 severity rating. A cracked ramp and an absent ramp are different budget lines.
- Pittsburgh sweep.** Run over un-audited street segments; fuse detections to per-segment scores.
- The deliverable.** Repair list ranked by severity × pedestrian demand (bus stops, schools, hospitals, senior housing), plus a map layer.

HOW WE EVALUATE

- Held-out city, not held-out images.** Train NYC/Portland/Bend → test on Pittsburgh's human labels. The real question is transfer.
- Per-class P/R at severity thresholds.** Reported separately — surface problems will be far worse than curb ramps.
- Against human disagreement**, not a fictional perfect label — labels carry agree / disagree / unsure votes from other volunteers, and that agreement is the ceiling.
- Audit-km saved at 90% recall** on severe barriers — the number that makes the pitch to the city.
- Equity audit** by neighbourhood income and race: a model that works in Shadyside and fails in Homewood misdirects the budget it steers.
- Human study:** 100 sampled predictions shown to 5–10 wheelchair / walker / stroller users.

DELIVERABLE LADDER – WHAT WE SHIP EVEN IF PITTSBURGH FAILS

TIER	WHAT WE CAN CLAIM	WHAT IT NEEDS	WHY IT CANNOT FAIL	DONE BY
0	Multi-class barrier classifier for all five types	Five 2 GB HF crop datasets, one GPU. No imagery acquisition at all.	Images ship with the data, splits are pre-defined, and published per-class models give us a comparison. Nothing external can break it.	Week 4
1	A curb-ramp detector trained on New York, Portland and Bend	rampnet-dataset (462 GB, streamed or subset) + rampnet-benchmark.	This is the answer to "what if Pittsburgh falls through". A fine-tuned detector scored on the same 1,000-panorama gold set as the published 0.949 / 0.873 — a complete, directly comparable result with zero Pittsburgh dependency.	Week 7
2	Pittsburgh repair-prioritization layer from existing human labels	The 25,690 API labels + WPRDC stops / demographics. Still no imagery.	Built entirely from data already collected: the ranking, map and equity analysis stand whether or not the model ever runs.	Week 9
3	Full model sweep of the un-audited 494 km	Tier 0+1 models + street imagery for ~5–10k Pittsburgh locations.	The only tier with an external dependency. Everything above already carries the project.	Week 12

BLOCKERS, LIMITATIONS AND WAYS THIS FAILS

		ORDERED BY HOW LIKELY THEY ARE TO ACTUALLY HURT · WEEK TO SETTLE EACH ONE	
Pittsburgh imagery — the real blocker The API gives labels, pixel coords and pano IDs — not pixels . Pittsburgh labels point at Google Street View, and bulk tile scraping (what the older tooling did) is ToS-questionable. → Official Street View Static API (10k free calls/mo), or Mapillary (free, CC-BY-SA — verify Pittsburgh coverage first). If both fail, fall back to Tiers 0–2.	WK 1	Transfer may simply fail Pittsburgh is hilly, its panoramas are a different vintage (labels average 767 days older than their imagery), and NYC/Portland/Bend look nothing like it. A large accuracy drop is likely. → Frame the drop as the result, not the failure — "state-of-the-art detection loses X points on an unseen city" is a legitimate finding here, if we decide in week 5 to report it that way.	WK 5
462 GB training set Can't be downloaded casually on campus wifi and won't sit on a laptop. → HF streaming=True, a subset of the 128 shards per split, or stage it in S3 beside the GPU. Cost is negligible; transfer time is not.	WK 2	Hard classes, noisy severity Obstacles and surface problems are rarer and far more subjective than curb ramps; severity SD is ~0.8 on a 1–5 scale, so exact-match severity will look terrible. → Report per class, never pooled; ordinal metrics (MAE, ±1 accuracy) for severity. Set that expectation in the proposal, not the final report.	WK 6
The "ground truth" is itself model-generated RampNet's 850k labels came from projecting government point inventories into panoramas with a crop model — 43% of records never yielded usable data . Training on it inherits its errors and blind spots. → Never headline a number computed on the auto-labels. Score only on the 1,000-panorama human benchmark and Pittsburgh's human labels.	WK 3	Biased evaluation set Volunteers audit where they live, so the 334 audited km are not a random sample of the 494 km we sweep. → Report per-neighbourhood coverage next to accuracy, and say plainly what the test set represents.	WK 8
		Stakeholders and scope If the grade depends on the City or Project Sidewalk answering email, one unanswered thread sinks it. Separately, the routing demo and live map are the fun parts and will eat the last three weeks. → Email both in week 2 but scope every deliverable to stand without a reply. Cut order if time runs short: routing demo → live map → severity head. Evaluation and the equity audit are never cut.	WK 2 / 10